This is an ambient light and proximity sensor module based on the Broadcom APDS-9930 chip, commonly used in smart devices for automatic brightness adjustment, object proximity detection, and low-power wake-up scenarios.

I. Core Parameters

Parameter Details
Power Supply Voltage: 2.2V~3.6V (3.3V recommended, do not connect to 5V)
Interface Type: I2C digital interface (default address 0x39)
Light Sensing Range: 0.01 lux ~ 100 klux (approaching human eye response curve)
Proximity Detection Distance: Typical value 5cm (up to 10cm without obstruction)
Operating Current: Ambient light mode: approx. 0.5mA
Proximity detection mode: approx. 2.5mA
Low power mode: approx. 0.1 μA
Operating Temperature: -40℃ ~ +85℃

II. Features
Dual Function Integration: Integrates an ambient light sensor (ALS) and an infrared proximity sensor within a single chip, achieving both detection functions without additional components.

High-sensitivity light sensing: Operates in extremely low light conditions (0.01 lux), supports detection through dark glass, and is suitable for automatic brightness adjustment in devices such as mobile phones and laptops.

Low-power design: Built-in state machine automatically enters low-power mode between detections, with average current in the microamp range, making it ideal for battery-powered devices.

Interrupt output: Supports light intensity threshold interrupts and proximity detection interrupts, which can directly trigger MCU wake-up, reducing polling power consumption.

III. Module Pin Definitions

Pin Function

VCC Power Input (3.3V)

GND Ground

SCL I2C Clock Line

SDA I2C Data Line

INT Interrupt Output Pin (Configurable Trigger Condition)

ULED Infrared LED Driver Output (The module integrates an infrared LED; this pin is for internal connection)

IV. Typical Application Scenarios

Smartphones/Tablets: Automatic screen brightness adjustment, screen off during calls to prevent accidental touches

Smart Home: Proximity sensor wake-up, automatic light switching

Wearable Devices: Heart rate monitoring module wake-up, raise-to-wake screen

Industrial Equipment: Object placement detection, low-power wake-up trigger

V. Arduino Wiring Example

Arduino Pins Module Pins

3.3V VCC

GND GND

A4 (SDA) SDA

A5 (SCL) SCL